

Tanzima Z. Islam

Associate Professor
Texas State University
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Education

- 2013 **Ph.D., Electrical and Computer Engineering**, *Purdue University*.
Thesis: Reliable and scalable checkpointing systems for distributed computing environments.
- 2006 **B.Sc., Computer Science & Engineering**, *Bangladesh University of Engineering & Technology*.

Professional Experience

- 2026–Present **Associate Computational Scientist**, *Brookhaven National Laboratory (BNL)*, AI Department, Computing and Data Sciences Directorate.
- 2025–Present **Associate Professor**, *Texas State University (TXST)*, Department of Computer Science.
- 2020–2025 **Assistant Professor**, *Texas State University (TXST)*, Department of Computer Science.
- Summer 2024 **Visiting Faculty**, *Oak Ridge National Laboratory (ORNL)*, Computer Science and Mathematics Division (CSMD).
- Summers 2022 and 2023 **Visiting Scholar**, *Brookhaven National Laboratory (BNL)*, Computational Science Initiative.
- Summer 2019 **Visiting Scholar**, *Lawrence Berkeley National Laboratory (LBNL)*, Computation Directorate.
- 2017–2019 **Assistant Professor**, *Western Washington University (WWU)*, Department of Computer Science.
- 2013–2017 **Postdoctoral Research Staff Member**, *Lawrence Livermore National Laboratory (LLNL)*, Center for Applied Scientific Computing.
Developed machine-learning techniques for performance analysis.
- 2006–2007 **Member, Research and Development**, *Commlink Info Tech Ltd.*, Dhaka, Bangladesh.
Developed software for a service-independent telecommunication network (Intelligent Network).

Awards, Honors & Inventions

- 2026 **ACM SIGHPC Emerging Woman Leader in Technical Computing Award**.
- 2026 **Inaugural Genesis Mission Phase I Award**, Department of Energy.
- 2025 **NSF CAREER Award**.
- 2025 **Best Paper Award**, IEEE Cloud Summit.
- 2025, 2024 **Research Millionaire Award**, College of Science and Engineering, Texas State University.
- 2025, 2023 **Presidential Distinction for Excellence in Scholarly/Creative Activities**, College of Science and Engineering, Texas State University.
- 2023 **Presidential Seminar Award for Excellence in Scholarly/Creative Activities**, Texas State University.
- 2023 **Valero Award for Excellence in Scholarly/Creative Activities**, Academic Affairs Council, Texas State University.
- 2022 **DOE Early Career Research Program (ECRP) Award**.
- 2021 **College of Science and Engineering Achievement Award for Excellence in Scholarly/Creative Activities**, Texas State University.
- 2019 **R&D 100 Award**, in collaboration with Lawrence Livermore National Laboratory for inventing a scalable software solution—the Scalable Checkpoint/Restart Framework 2.0 (SCR)—that makes High Performance Computing Systems reliable and efficient.

- 2015, 2014 Best Poster Award, LLNL Annual Scholars Poster Symposium.
- 2014 **Lawrence Livermore National Laboratory (LLNL) Director's Science & Technology Award for Excellence in Publication.**
- 2014 Second Place Winner, LLNL Computation Directorate Postdoctoral Poster Symposium.
- 2012, 2009 Best Student Paper Finalist, International Conference for High Performance Computing, Networking, Storage and Analysis (SC).
- 2010 Second Place Winner, ACM Student Research Competition.

Research and Other Funding

Secured \$5.65M+ total in external funding as PI+Co-PI (PI only: \$3.535M+).

- 2026–2027 **PI**, OPTIX: A Multi-Modal Foundation Model for Performance-Aware HPC Code Intelligence, Department of Energy's Inaugural Genesis Mission Phase I Award, \$750K (9 months, TXST: \$300K). Collaborators: Brookhaven National Laboratory, Lawrence Livermore National Laboratory, University of North Carolina at Chapel Hill (5% acceptance rate).
- 2026 **PI**, NSF REU Supplemental Award, CRA, \$10K.
- 2025–2030 **PI**, CAREER: SPEED: Scalable Performance-aware Engine for Efficient Decision-making on HPC Systems, National Science Foundation (NSF) CAREER program, \$650K.
- Summer 2024 **PI**, Application Fingerprinting, Department of Energy's STEP: Software Tools Ecosystem Project at Oak Ridge National Laboratory, \$16K.
- Summer 2023 **PI**, SRP-HPC: Performance Analysis using Machine Learning: Use Case Chimbuko, DOE Sustainable Research Pathways (SRP) Fellowship, \$85K.
- 2023–2026 **Sub-contract**, FRACTAL-SI: Bridging Source-Target Feature Gaps for Generalizable HPC Performance Prediction, Laboratory Directed Research and Development (LDRD) at Lawrence Livermore National Laboratory, \$310K.
- 2022–2027 **PI**, INTELYTICS: An Efficient Data-Driven Decision-Making Engine for Performance in the Era of Heterogeneity, DOE Early Career Research Program (ECRP), \$770K (10% acceptance rate).
- 2022–2026 **Co-PI**, RECUP: Scalable Metadata and Provenance Services for Reproducible Hybrid Workflows, DOE Next Generation Data Management, \$2.43M (TXST PI: \$300K).
- 2022–2023 **Sub-contract**, ICE4HPC: Towards the Intelligent Center for HPC, LDRD at Lawrence Livermore National Laboratory, \$100K.
- 2022 **PI**, SRP-HPC: Characterizing Workflow Applications using Machine Learning, DOE SRP Fellowship at Brookhaven National Laboratory, \$68K.
- 2021–Present **PI**, PerfROCm: Cross-platform performance prediction and analysis using deep learning, AMD Research Gift (renewed annually since 2021), \$225K.
- 2021–2024 **Member**, REU Site: Research Experiences for Undergraduates in Edge Computing, NSF, \$389K.
- 2021 **PI**, Predicting Performance using Few-Shot Learning, Research Enhancement Program (REP) at Texas State University, \$8K.
- 2021 **Co-PI**, AMD COVID HPC Grant—equipment grant for a 5-petaflops HPC system, \$400K (estimated value).
- Summer 2019 **PI**, Proxy Application Validation for Exascale Co-design, DOE SRP Fellowship at Lawrence Berkeley National Laboratory, \$43K.
- 2018 **Co-PI**, Scientific Data Visualization course development, Office of Research and Sponsored Programs at Western Washington University, \$12K.
- 2017–2022 **PI**, Parallel Computing course, Time allocation grant from XSEDE. 100K core-hours.
- 2016 **PI**, VERITAS for Understanding Performance Evolution during Code Development, Linking Exploratory Application Research to Next-gen Development at Lawrence Livermore National Laboratory, \$200K.

Research Publications

Papers

- [1] Mohammad Zaeed, **Tanzima Z. Islam**, and Vladimir Indic. [OPTIMAS: An Intelligent Analytics-Informed Generative AI Framework for Performance Optimization](#). In *Proceedings of the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.2*, KDD '26, pages 6373–6384, New York, NY, USA, August 2026. Association for Computing Machinery. Acceptance rate: 18.5%.
- [2] Elvis G. Fefey and **Tanzima Z. Islam**. [Load-Aware Offline Scheduling for Heterogeneous Edge Deep Learning Inference](#). In *2026 IEEE Cloud Summit*, pages 98–103, June 2026. Acceptance rate: 34.4%.
- [3] Awatif Yasmin, Tarek Mahmud, Sana Alamgeer, **Tanzima Z. Islam**, and Anne H. H. Ngu. [MODELX_c: A Novel Data-Efficient Personalization for Fall Detection](#). In *2026 IEEE International Conference on Digital Health (ICDH)*, pages 261–271, May 2026. Acceptance rate: 32.4%.
- [4] **Tanzima Z. Islam**, Holland Schutte, Mohammad Zaeed, and Aniruddha Marathe. [Data-driven analysis to understand GPU hardware resource usage of optimizations](#). *The International Journal of High Performance Computing Applications (IJHPCA)*, 40(3):404–420, April 2026.
- [5] Ashna Nawar Ahmed, Banooqa H. Banday, Terry Jones, and **Tanzima Z. Islam**. [Attention-Informed Surrogates for Navigating Power-Performance Trade-offs in HPC](#). In *Workshop on ML for Systems in Conjunction with NeurIPS*, 2025.
- [6] Arunavo Dey, Neil Antony, Aakash Raj Dhakal, Kowshik Thopalli, Jayaraman J. Thiagarajan, Tapasya Patki, Aniruddha Marathe, Tom Scogland, Jae-Seung Yeom, and **Tanzima Z. Islam**. [MODELX: A Novel Transfer Learning Approach Across Heterogeneous Datasets](#). In *Proceedings of the 34th International Symposium on High-Performance Parallel and Distributed Computing*, HPDC '25, New York, NY, USA, 2025. Association for Computing Machinery. Acceptance rate: 19%.
- [7] Aakash Raj Dhakal, **Tanzima Z. Islam**, Arunavo Dey, Daniel Nichols, Abhinav Bhatele, Tapasya Patki, Tom Scogland, and Jae-Seung Yeom. [xAMM: "Attention" to Details Improves Cross-Platform Prediction Accuracy](#). In *2025 IEEE 25th International Symposium on Cluster, Cloud and Internet Computing (CCGrid)*, pages 1–10, 2025. Acceptance rate: 25%.
- * [8] Elvis G. Fefey and **Tanzima Z. Islam**. [Optimizing Deep Learning Inference on Heterogeneous Devices on the Edge: An ILP-Based Rate Monotonic Scheduling Approach](#). In *2025 IEEE Cloud Summit*, pages 23–28, 2025. **Best Paper Award**.
- [9] Christopher Kelly, Wei Xu, Line C. Pouchard, Hubertus Van Dam, **Tanzima Z. Islam**, Shinjae Yoo, and Kerstin Kleese Van Dam. [Performance Analysis and Data Reduction for Exascale Scientific Workflows](#). *The International Journal of High Performance Computing Applications (IJHPCA)*, 2025.
- [10] Matthias Maiterth, Wesley H. Brewer, Jaya S. Kuruvella, Arunavo Dey, **Tanzima Z. Islam**, Rashadul Kabir, Kevin Menear, Dmitry Duplyakin, Tapasya Patki, Terry Jones, and Feiyi Wang. [HPC Digital Twins for Evaluating Scheduling Policies, Incentive Structures and their Impact on Power and Cooling](#). In *Proceedings of the SC '25 Workshops of the International Conference for High Performance Computing, Networking, Storage and Analysis*, SC Workshops '25, pages 1959–1969, New York, NY, USA, 2025. Association for Computing Machinery.
- [11] Tarek Ramadan, Nathan Pinnow, Chase Phelps, Jayaraman J. Thiagarajan, and **Tanzima Z. Islam**. [Structure-Aware Representation Learning for Effective Performance Prediction](#). *Concurrency and Computation: Practice and Experience*, 37(9-11):e70046, 2025.
- [12] Raúl Sirvent, Rocío Carratalá-Saez, Amal Gueroudji, **Tanzima Z. Islam**, Line C. Pouchard, and Michela Tauber. [Reproducibility for HPC and Distributed Environments: Committees, Nondeterminism, Performance and Workflows](#). In *ACM Conference on Reproducibility and Replicability (ACM REP)*, ACM REP '25, New York, NY, USA, 2025. Association for Computing Machinery.
- [13] Banooqa H. Banday, Kowshik Thopalli, **Tanzima Z. Islam**, and Jayaraman J. Thiagarajan. [On the Role of Prompt Construction in Enhancing Efficacy and Efficiency of LLM-Based Tabular Data Generation](#). In *2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, April 2025.
- [14] Banooqa H. Banday, **Tanzima Z. Islam**, and Aniruddha Marathe. [PERFGEN: A Synthesis and Evaluation Framework for Performance Data using Generative AI](#). In *2024 IEEE 48th Annual Computers, Software, and Applications Conference (COMPSAC)*, pages 188–197, 2024. Acceptance rate: 24%.
- [15] Arunavo Dey, Aakash Raj Dhakal, **Tanzima Z. Islam**, Jae-Seung Yeom, Tapasya Patki, Daniel Nichols, Alexander Movsesyan, and Abhinav Bhatele. [Relative Performance Prediction Using Few-Shot Learning](#). In *2024 IEEE 48th Annual Computers, Software, and Applications Conference (COMPSAC)*, pages 1764–1769, 2024. Acceptance rate: 24%.

- [16] Elvis G. Fefey and **Tanzima Z. Islam**. [Toward Efficient Deep Learning Inference: On-Node Heterogeneous Scheduling in Edge-Cloud Infrastructure](#). In *2024 IEEE Cloud Summit*, pages 73–78, 2024.
- [17] Amal Gueroudji, Chase Phelps, **Tanzima Z. Islam**, Philip Carns, Shane Snyder, Matthieu Dorier, Robert B. Ross, and Line C. Pouchard. [Performance Characterization and Provenance of Distributed Task-based Workflows on HPC Platforms](#). In *Proceedings of the SC '24 Workshops of the International Conference for High Performance Computing, Networking, Storage and Analysis, SC-W '24*, pages 2032–2039. IEEE Press, 2024.
- [18] Chase Phelps, Ankur Lahiry, **Tanzima Z. Islam**, and Line C. Pouchard. [Reimagine Application Performance as a Graph: Novel Graph-Based Method for Performance Anomaly Classification in High-Performance Computing](#). In *2024 IEEE 48th Annual Computers, Software, and Applications Conference (COMPSAC)*. IEEE, 2024. Acceptance rate: 24%.
- [19] Mohammad Zaeed, **Tanzima Z. Islam**, and Vladimir Indic. [Characterize and Compare the Performance of Deep Learning Optimizers in Recurrent Neural Network Architectures](#). In *2024 IEEE 48th Annual Computers, Software, and Applications Conference (COMPSAC)*, pages 39–44, 2024. Acceptance rate: 24%.
- [20] Arunavo Dey, **Tanzima Z. Islam**, Chase Phelps, and Christopher Kelly. [Signal Processing Based Method for Real-Time Anomaly Detection in High-Performance Computing](#). In *2023 IEEE 47th Annual Computers, Software, and Applications Conference (COMPSAC)*, pages 233–240, 2023. Acceptance rate: 27%.
- [21] Bogdan Nicolae, **Tanzima Z. Islam**, Robert B. Ross, Hubertus Van Dam, Kevin Assogba, Polina Shpilker, Mikhail Titov, Matteo Turilli, Tianle Wang, Ozgur O. Kilic, Shantenu Jha, and Line C. Pouchard. [Building the I \(Interoperability\) of FAIR for Performance Reproducibility of Large-Scale Composable Workflows in RECUP](#). In *2023 IEEE 19th International Conference on e-Science (e-Science)*, pages 1–7, 2023.
- [22] Chase Phelps and **Tanzima Z. Islam**. [Automatic Parallelization of Cellular Automata for Heterogeneous Platforms](#). In *2023 IEEE 47th Annual Computers, Software, and Applications Conference (COMPSAC)*, pages 352–361, 2023. Acceptance rate: 27%.
- [23] Tarek Ramadan, Ankur Lahiry, and **Tanzima Z. Islam**. [Novel Representation Learning Technique Using Graphs for Performance Analytics](#). In *2023 International Conference on Machine Learning and Applications (ICMLA)*, pages 1311–1318, 2023.
- [24] Line C. Pouchard, **Tanzima Z. Islam**, and Bogdan Nicolae. [Challenges for Implementing FAIR Digital Objects with High Performance Workflows](#). *Research Ideas and Outcomes*, 8:e94835, 2022.
- [25] Holland Schutte, Chase Phelps, Aniruddha Marathe, and **Tanzima Z. Islam**. [libNVCD: An Extendable and User-friendly Multi-GPU Performance Measurement Tool](#). In *2022 IEEE 46th Annual Computers, Software, and Applications Conference (COMPSAC)*, pages 73–82, 2022.
- [26] **Tanzima Z. Islam** and Chase Phelps. [HPC@SCALE: A Hands-on Approach for Training Next-Gen HPC Software Architects](#). In *2021 IEEE 28th International Conference on High Performance Computing, Data and Analytics Workshop (HiPCW)*, pages 29–34, 2021.
- [27] **Tanzima Z. Islam**, Philip Wu Liang, Forest Sweeney, Cody Pragner, Jayaraman J. Thiagarajan, Moushumi Sharmin, and Shameem Ahmed. [College Life is Hard!—Shedding Light on Stress Prediction for Autistic College Students using Data-Driven Analysis](#). In *2021 IEEE 45th Annual Computers, Software, and Applications Conference (COMPSAC)*, pages 428–437, 2021.
- [28] Quentin Jensen, Filip Jagodzinski, and **Tanzima Z. Islam**. [FILCIO: Application Agnostic I/O Aggregation to Scale Scientific Workflows](#). In *2021 IEEE 45th Annual Computers, Software, and Applications Conference (COMPSAC)*, pages 1587–1592, 2021.
- [29] Tarek Ramadan, **Tanzima Z. Islam**, Chase Phelps, Nathan Pinnow, and Jayaraman J. Thiagarajan. [Comparative Code Structure Analysis using Deep Learning for Performance Prediction](#). In *2021 IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS)*, pages 151–161, 2021. Acceptance rate: 36%.
- [30] Jack Stratton, Michael Albert, Quentin Jensen, Max Ismailov, Filip Jagodzinski, and **Tanzima Z. Islam**. [Towards Aggregation Based I/O Optimization for Scaling Bioinformatics Applications](#). In *2020 IEEE 44th Annual Computers, Software, and Applications Conference (COMPSAC)*, pages 1250–1255, 2020.
- [31] **Tanzima Z. Islam**, Alexis Ayala, Quentin Jensen, and Khaled Ibrahim. [Toward a Programmable Analysis and Visualization Framework for Interactive Performance Analytics](#). In *2019 IEEE/ACM International Workshop on Programming and Performance Visualization Tools (ProTools)*, pages 70–77, 2019.
- [32] Tapasya Patki, Jayaraman J. Thiagarajan, Alexis Ayala, and **Tanzima Z. Islam**. [Performance optimality or reproducibility: that is the question](#). In *Proceedings of the International Conference for High Performance*

Computing, Networking, Storage and Analysis (SC), SC '19, New York, NY, USA, 2019. Association for Computing Machinery. Acceptance rate: 20%.

- [33] Nicholas Majeske, Filip Jagodzinski, Brian Hutchinson, and **Tanzima Z. Islam**. [Low Rank Smoothed Sampling Methods for Identifying Impactful Pairwise Mutations](#). In *Proceedings of the 2018 ACM International Conference on Bioinformatics, Computational Biology, and Health Informatics, BCB '18*, pages 681–686, New York, NY, USA, 2018. Association for Computing Machinery.
 - [34] Jayaraman J. Thiagarajan, Rushil Anirudh, Bhavya Kailkhura, Nikhil Jain, **Tanzima Z. Islam**, Abhinav Bhatele, Jae-Seung Yeom, and Todd Gamblin. [PADDLE: Performance Analysis using a Data-driven Learning Environment](#). In *2018 IEEE International Parallel and Distributed Processing Symposium (IPDPS)*, May 2018. Acceptance rate: 24.5%.
 - [35] Tania Banerjee, Jason Hackl, Mrugesh Shringarpure, **Tanzima Z. Islam**, S. Balachandar, Thomas Jackson, and Sanjay Ranka. [A New Proxy Application for Compressible Multiphase Turbulent Flows](#). *Sustainable Computing: Informatics and Systems*, 16:11–24, 2017.
 - [36] Teng Wang, Adam Moody, Yue Zhu, Kathryn Mohror, Kento Sato, **Tanzima Z. Islam**, and Weikuan Yu. [MetaKV: A Key-Value Store for Metadata Management of Distributed Burst Buffers](#). In *2017 IEEE International Parallel and Distributed Processing Symposium (IPDPS)*, pages 1174–1183, 2017.
 - [37] **Tanzima Z. Islam**, Kathryn Mohror, and Martin Schulz. [Exploring the MPI Tool Information Interface: Features and Capabilities](#). *The International Journal of High Performance Computing Applications (IJHPCA)*, 30(2):212–222, 2016.
 - [38] Tania Banerjee, Jason Hackl, Mrugesh Shringarpure, **Tanzima Z. Islam**, Sivaramakrishnan Balachandar, T. Jackson, and Sanjay Ranka. [CMT-Bone—A Proxy Application for Compressible Multiphase Turbulent Flows](#). In *IEEE International Conference on High Performance Computing, Data, and Analytics (HiPC)*, pages 173–182. IEEE, December 2016.
 - [39] **Tanzima Z. Islam**, Jayaraman J. Thiagarajan, Abhinav Bhatele, Martin Schulz, and Todd Gamblin. [A Machine-Learning Framework for Performance Coverage Analysis of Proxy Applications](#). In *International Conference for High Performance Computing, Networking, Storage and Analysis (SC)*, November 2016. Acceptance rate: 23%.
 - [40] Lee Savoie, David K. Lowenthal, Bronis R. de Supinski, **Tanzima Z. Islam**, Kathryn Mohror, Barry Rountree, and Martin Schulz. [I/O Aware Power Shifting](#). In *2016 IEEE International Parallel and Distributed Processing Symposium (IPDPS)*, pages 740–749, May 2016. Acceptance rate: 23%.
 - [41] **Tanzima Z. Islam**, Kathryn Mohror, and Martin Schulz. [Exploring the Capabilities of the New MPI_T Interface](#). In *Proceedings of the 21st European MPI Users' Group Meeting, EuroMPI/ASIA '14*, pages 91–96, New York, NY, USA, 2014. Association for Computing Machinery.
 - [42] Anup Mohan, Thomas Hacker, Gregory P. Rodgers, and **Tanzima Z. Islam**. [Batchsubmit: A high-volume Batch Submission System for Earthquake Engineering Simulation](#). *Concurrency and Computation: Practice and Experience*, 26(13):2240–2252, 2014.
 - [43] John R. Tramm, Andrew R. Siegel, **Tanzima Z. Islam**, and Martin Schulz. [XSBench: The Development and Verification of a Performance Abstraction for Monte Carlo Reactor Analysis](#). In *PHYSOR 2014 - The Role of Reactor Physics toward a Sustainable Future*, 2014.
 - [44] **Tanzima Z. Islam**, Saurabh Bagchi, and Rudolf Eigenmann. [Reliable and Efficient Distributed Checkpointing System for Grid Environments](#). *Journal of Grid Computing (JoGC)*, 12(4):593–613, December 2014.
 - [45] **Tanzima Z. Islam**, Kathryn Mohror, Saurabh Bagchi, Adam Moody, Bronis R. de Supinski, and Rudolf Eigenmann. [mcrEngine: A Scalable Checkpointing System Using Data-Aware Aggregation and Compression](#). *Scientific Programming*, 21(3-4):149–163, 2013.
 - * [46] **Tanzima Z. Islam**, Kathryn Mohror, Saurabh Bagchi, Adam Moody, Bronis R. de Supinski, and Rudolf Eigenmann. [mcrEngine: A Scalable Checkpointing System Using Data-Aware Aggregation and Compression](#). In *International Conference for High Performance Computing, Networking, Storage and Analysis (SC)*, pages 17:1–17:11, 2012. Acceptance rate: 20%. **Best Student Paper Finalist**.
 - * [47] **Tanzima Z. Islam**, Saurabh Bagchi, and Rudolf Eigenmann. [FALCON: A System for Reliable Checkpoint Recovery in Shared Grid Environments](#). In *International Conference for High Performance Computing, Networking, Storage and Analysis (SC)*, pages 1–12, November 2009. **Best Student Paper Finalist**.
- Pre-prints, Manuscripts Under Preparation**
- [48] Ashna Nawar Ahmed and **Tanzima Z. Islam**. [AGENTRULEZ: An Application-Fingerprinting Agent for HPC Performance Telemetry](#). Under preparation.

- [49] Ashna Nawar Ahmed, Jaya Sravani Kuruvella, and **Tanzima Z. Islam**. Symbolic AI-Guided Power-Aware Scheduling Across HPC Workloads. Under preparation.
- [50] Arunavo Dey, Ankur Lahiry, Nawshin Tabassum, and **Tanzima Z. Islam**. MODELS: A Causal Graph-Based Embedding Learning Methodology for Effective Knowledge Transfer Across Heterogeneous HPC Datasets. Under preparation.
- [51] Aakash Dhakal and **Tanzima Z. Islam**. REGENT: Generation Without Retraining via Few-Shot Adaptation for Heterogeneous Tabular Domains. Under preparation.
- [52] Nawshin Tabassum, Sana Alamgeer, and **Tanzima Z. Islam**. Multi-Modal Data Fusion Improves Performance Prediction on HPC Systems. Under preparation.
- [53] Mohammad Zaeed and **Tanzima Z. Islam**. Prompting Is All You Need to Optimize Code. Under preparation.
- [54] Ankur Lahiry, Banooqa H. Bandy, Yugesh Bhattarai, Mohammad Zaeed, and **Tanzima Z. Islam**. [COMPASS: A Unified Decision-Intelligence System for Navigating Performance Trade-off in HPC](#). arXiv:2604.22688, April 2026.

Policy Reports, Book Chapters, Articles

- [55] Lois Curfman McInnes, Dorian Arnold, Prasanna Balaprakash, Mike Bernhardt, Franck Cappello, Beth Cerny, Deborah DiazGranados, Anshu Dubey, Nichole Etienne, Roscoe Giles, Diego Gomez-Zara, Denice Ward Hood, Mary Ann Leung, Vanessa Lopez-Marrero, Olivia B. Newton, Irene Qualters, Keita Teranishi, Stefan M. Wild, Gabrielle Allen, Richard Arthur, Alexandra Ballow, Tony Baylis, David E. Bernholdt, Daniel Bielich, Johanna Cohoon, Jeremy Crampton, Charles Ferenbaugh, Stephen M. Fiore, Thomas Herault, **Tanzima Z. Islam**, Stephen Jacobsohn, Meifeng Lin, Charles Lively, Satoshi Matsuoka, Stasa Milojevic, Daniel Nichols, Chris Oehmen, Santiago Ospina Tabares, Michael E. Papka, Katherine Riley, Damian Rouson, Sudip K. Seal, Brittany Segundo, John Shalf, Andrew R. Siegel, Valerie Taylor, Jim Willenbring, and Lou Woodley. [Report of the 2026 Workshop on Next-Generation Ecosystems for Scientific Computing: Harnessing Community, Software, and AI for Cross-Disciplinary Team Science](#). arXiv:2608.26519, 2026.
- [56] Lois Curfman McInnes, Dorian Arnold, Prasanna Balaprakash, Mike Bernhardt, Beth Cerny, Anshu Dubey, Roscoe Giles, Denice Ward Hood, Mary Ann Leung, Vanessa Lopez-Marrero, Paul Messina, Olivia B. Newton, Chris Oehmen, Stefan M. Wild, Jim Willenbring, Lou Woodley, Tony Baylis, David E. Bernholdt, Chris Camano, Johanna Cohoon, Charles Ferenbaugh, Stephen M. Fiore, Sandra Gesing, Diego Gomez-Zara, James Howison, **Tanzima Z. Islam**, David Kepczynski, Charles Lively, Harshitha Menon, Bronson Messer, Marieme Ngom, Umesh Paliath, Michael E. Papka, Irene Qualters, Elaine M. Raybourn, Katherine Riley, Paulina Rodriguez, Damian Rouson, Michelle Schwalbe, Sudip K. Seal, Ozge Surer, Valerie Taylor, and Lingfei Wu. [Report of the 2025 Workshop on Next-Generation Ecosystems for Scientific Computing: Harnessing Community, Software, and AI for Cross-Disciplinary Team Science](#). arXiv:2510.03413, 2025.
- [57] SIAM Task Force on the Future of Computational Science. [SIAM Task Force Anticipates Future Directions of Computational Science](#). *SIAM News*, June 2024.
- [58] Chris Daw, Brian Barragan Cruz, Nicholas Majeske, Filip Jagodzinski, **Tanzima Z. Islam**, and Brian Hutchinson. Low rank approximation methods for identifying impactful pairwise protein mutations. In Nurit Haspel, Filip Jagodzinski, and Kevin Molloy, editors, *Algorithms and Methods in Structural Bioinformatics*, pages 63–87. Springer Nature, 2022.
- [59] Ritu Arora, Xiaosong Li, Bonnie Hurwitz, Daniel Fay, Dhabaleswar K. Panda, Edward Valeev, Shaowen Wang, Shirley Moore, Sunita Chandrasekaran, Ting Cao, Holly Bik, Matthew Curry, and **Tanzima Z. Islam**. [Future Directions of the Cyberinfrastructure for Sustained Scientific Innovation \(CSSI\) Program](#). arXiv:2010.15584, 2020.

Ph.D. Dissertation

- [60] **Tanzima Z. Islam**. [Reliable and scalable checkpointing systems for distributed computing environments](#). PhD thesis, Purdue University, West Lafayette, IN, May 2013.

Research Posters and Abstracts

- * [61] Yugesh Bhattarai, Jessica Rubin, Prajan Kumar Manoj Rekha, Mohammad Zaeed, and **Tanzima Z. Islam**. OPTIMASX: An Agentic End-to-End GPU Performance Optimization Workflow in the Code Editor. In *International Conference for High Performance Computing, Networking, Storage and Analysis (SC'26)*, November 2026. [Best Research Poster Award Finalist](#).
- [62] Ankur Lahiry and **Tanzima Z. Islam**. What Decisions Should HPC Make So You Don't Have To? In *New York Scientific Data Summit (NYSDS'26)*, September 2026.

- [63] Mohammad Zaeed and **Tanzima Z. Islam**. From Hardware Characterization to AI-Guided Optimization: A Unified Framework for Automated HPC Performance Engineering. In *14th Annual MVAICH User Group Conference (MUG'26)*, August 2026.
- [64] Arunavo Dey, Jae-Seung Yeom, Tapasya Patki, and **Tanzima Z. Islam**. **QUANTUMX: A Few-Shot-Learning Approach for Quantum Performance Modeling in HPC**. In *International Workshop on Software Frameworks for Integrating Quantum and HPC Ecosystems, in conjunction with ICS'25*, 2025.
- [65] Ashna Nawar Ahmed, Banooqa H. Banday, Terry Jones, and **Tanzima Z. Islam**. Intelligent Surrogates Pay Attention to Data, Improving Multi-Objective HPC Optimization. In *International Conference for High Performance Computing, Networking, Storage and Analysis (SC'25)*, November 2025.
- [66] Chase Phelps, Ankur Lahiry, **Tanzima Z. Islam**, and Christopher Kelly. Graph Based Anomaly Detection in Chimbuko: Feasible or Fallible? In *International Conference for High Performance Computing, Networking, Storage and Analysis (SC'23)*, November 2023.
- [67] Alicia Guite, **Tanzima Z. Islam**, Christopher Kelly, and Wei Xu. Interactive Visual Analysis Tool for Anomaly Provenance Data. In *International Conference for High Performance Computing, Networking, Storage and Analysis (SC'22)*, November 2022.
- [68] Mohammad Zaeed, **Tanzima Z. Islam**, Younghyun Cho, Xiaoye S. Li, Hengrui Luo, and Yang Liu. Analysis and Visualization of Important Performance Counters to Enhance Interpretability of Autotuner Output. In *International Conference for High Performance Computing, Networking, Storage and Analysis (SC'22)*, November 2022.
- [69] Logan Moody, Nathan Pinnow, Michael Lam, Harshitha Menon, Markus Schordan, Scott G. Lloyd, and **Tanzima Z. Islam**. Automatic Generation of Mixed-Precision Programs. In *International Conference for High Performance Computing, Networking, Storage and Analysis (SC'18)*, November 2018.
- * [70] Konstantinos Chasapis and **Tanzima Z. Islam**. Toward Scientific-Data Compression Using Variable Clustering. In *LLNL Annual Scholars Poster Symposium*, August 2015. **Best Poster Award**.
- [71] Simone Smarr, **Tanzima Z. Islam**, and Yolanda Rankin. Modular Extensible Framework for Performance Comparative Analysis. In *ACM Richard Tapia Celebration of Diversity in Computing Conference*, February 2015.
- * [72] Xiang Ni, **Tanzima Z. Islam**, Kathryn Mohror, Adam Moody, and Laxmikant V. Kale. Lossy Compression for Checkpointing: Fallible or Feasible? In *International Conference for High Performance Computing, Networking, Storage and Analysis (SC'14)*, November 2014. **Best Poster Award** at 2014 LLNL Annual Scholars Poster Symposium.
- [73] Mohammad S. Hossain, **Tanzima Z. Islam**, Jinliang Wei, Vijay Raghunathan, and Saurabh Bagchi. **EUNOMIA: A Fast and Collaborative Interference Avoidance Technique for Wireless Medical Devices**. Technical Report 422, Purdue University, Department of Electrical and Computer Engineering, September 2011.
- [74] **Tanzima Z. Islam**, Kathryn Mohror, Adam Moody, Bronis R. de Supinski, Saurabh Bagchi, and Rudolf Eigenmann. Data-Aware Inter-Process Checkpoint Compression. In *International Conference for High Performance Computing, Networking, Storage and Analysis (SC'10)*, November 2010.
- * [75] **Tanzima Z. Islam**, Saurabh Bagchi, and Rudolf Eigenmann. Harnessing Multiple Cores for Efficient Checkpointing in Grid Systems. In *ACM Student Research Competition (SRC) at Anita Borg Institute Grace Hopper Celebration of Women in Computing*, October 2010. **ACM Student Research Competition Second Place Winner**.

Selected Open-Source Software & Dataset Releases

- GPTune** Integrated the interpretable machine learning algorithm for identifying important performance-influencing factors—developed as part of the Dashing framework—into GPTune, an auto-tuner developed at Lawrence Berkeley National Laboratory.
- SCR** Integrated a scalable data compression method—mcrEngine—into Scalable Checkpoint/Restart library (SCR) for MPI applications in collaboration with LLNL. SCR will serve as the reliability backbone for LLNL's exascale system, El Capitan.
- libNVCD** Easy-to-use, performance measurement and analysis tool for NVIDIA based GPUs.
- Dashing** Interpretable machine learning algorithms for performance characterization and comparison.
- Gyan** Performance measurement tool for MPI libraries in collaboration with LLNL.
- PyPerfDump** Hardware performance counter measurement tool for Python applications.
- Dataset 1** On-node scaling data on HPC systems. [doi:10.5281/zenodo.4315003](https://doi.org/10.5281/zenodo.4315003).

- Dataset 2** Performance characterization data for several DOE Exascale Computing Project (ECP) applications including Particle-In-Cell application WarpX and cosmological simulation code Nyx3D. [doi:10.5281/zenodo.3403037](https://doi.org/10.5281/zenodo.3403037).
- Dataset 3** OPTIMAS Dataset: Prompts, LLM Responses, and Runtime Measurements for Diagnostic-Informed GPU Code Optimization. [doi:10.5281/zenodo.20499115](https://doi.org/10.5281/zenodo.20499115).

Selected Professional Activities

Chair	General Co-chair@IPDPS'27, Student Program Co-chair@SC'27, Posters@ISC'26, Performance Track@SC'26, Software Track@ICPP'26, Performance Track@IPDPS'25, Applications and Workflows Track@CCGrid'25, Posters@HPDC'25, Performance Track@SC'22.
Organizer	IEEE International Workshop on Big Data Computation, Analysis & Applications (in conjunction with IEEE COMPSAC'19–'23).
Vice/Co-Chair	Posters@ISC'25, Performance Track@HiPC'24, Awards@SC'23, Performance Track@IEEE Cluster'23, Performance Track@ICPP'19.
Selected Technical Program Committees	NeurIPS'22, '26; SIGKDD'26; High-Performance Distributed Computing (HPDC)'22–Present; International Parallel & Distributed Processing Symposium (IPDPS)'17–Present; International Conference for High Performance Computing, Networking, Storage and Analysis (SC)'17–Present (Performance track); SC'19–'21 (Posters); SC'18 (HPC for Undergraduates); SC'15 (Resilience BoF); International Supercomputing Conference (ISC)'19–Present; Women in HPC Technical Conference'20; EuroMPI User Forum'20; Grace Hopper Celebration (GHC)'19 (Computer Systems Engineering track); The Platform for Advanced Scientific Computing (PASC)'19; International Conference on Parallel Processing (ICPP)'18–Present; International Conference on Distributed Computing Systems (ICDCS)'17 (Workshop); International Symposium on Computer Architecture and High Performance Computing (SBAC-PAD)'13–'14; Symposium on Principles and Practice of Parallel Programming (PPoPP)'11; Dependable Systems and Networks (DSN)'10, '22.
Editor	Subject matter editor for Future Generation Computer Systems (FGCS)'25; Parallel Computing (ParCo)'25–Present.
Advisor/Mentor	CAHSI NAIRR Grant Program.
Task Force	Member, SIAM Task Force on the Future of Computational Science, Washington, D.C.'23.
Workshops	Attendee, NSF/DOE Workshop on NAIRR Software, Lemont, IL'25.
Technical Board	Technical review board member at IEEE Transactions on Parallel and Distributed Systems (TPDS)'20–Present.
Grant Review Committees	DOE ECRP; DOE CSGF'25–'26; DOE SciDAC; DOE SBIR/STTR; NSF FIRE; NSF ACSS (Category II); NSF Cybertraining/OAC; NSF CISE/CSSI; NSF CISE/SHF; NSF CISE/OAC; NSF CISE/CCF; NSF HBCU-UP.
Selected Journal Reviewing	IEEE Transactions on Parallel and Distributed Systems (TPDS)'19–Present; Parallel Computing (ParCo)'19–'21; Journal of Grid Computing (JoGC)'17–'21; International Journal of High Performance Computing Applications (IJHPCA)'18; Concurrency and Computation: Practice and Experience (CCPE)'16–'17.
Memberships	IEEE , ACM .

Invited Talks and Panels

September, 2026	What Decisions Should HPC Make So You Do Not Have To? , <i>CCDSC</i> , Lyon, France, Invited by Turing Award winner Dr. Jack Dongarra.
July, 2026	AI4Performance: System Intelligence as a Service for the Post-Exascale Era , <i>AI Department, Brookhaven National Laboratory (BNL)</i> , Virtual.
June, 2026	From Modeling to Decision Intelligence: Learning to Navigate Power–Performance Trade-offs in HPC Systems , <i>Women in HPC (WHPC) at the International Supercomputing Conference (ISC)</i> , Hamburg, Germany.
June and August, 2025	From Reactive Debugging to Proactive Detection: AI/ML for Performance-Aware Software Development , <i>Monterey Data Conference (invite-only DOE workshop); PASC; EuroPar</i> , Monterey, CA; Switzerland; Germany.
2025, 2024	INTELYTICS: An Efficient Data-Driven Decision-Making Engine for Performance in the Era of Heterogeneity , <i>DOE ASCR PI Meeting</i> , Frisco, TX and Atlanta, GA, respectively.
November, 2024	Women in HPC Panel , <i>San Diego Supercomputer Center panel at the International Conference for High Performance Computing, Networking, Storage and Analysis (SC)</i> , Atlanta, GA.

- September, 2024 **Reimagine Performance Modeling: Challenges and Opportunities with Generative AI**, *CCDSC*, Lyon, France, Invited by Turing Award winner Dr. Jack Dongarra.
- August, 2024 **Augmenting AMD's Omniperv with ML-based In-depth Influence Analysis for Explaining Performance Gap**, *CORAL-2 Meeting at Lawrence Livermore National Laboratory (LLNL)*, Virtual.
- 2024 **Reimagine Performance Modeling: Challenges and Opportunities with Generative AI (expanded)**, *GCLab, University of Tennessee*, Knoxville, TN.
- 2023 **Building Your Research Brand**, *Rising Stars Panel at the University of Texas at Austin*, Austin, TX.
- 2023 **Anomaly Detection, Generation, and Offline Analysis Capabilities in Chimbuko**, *Sustainable Horizons Institute (SHI)*.
- October, 2022 **Characterization of Deep Learning Inference Workloads**, *New York Scientific Data Summit (NYSDS)*, Virtual, Lightning talk presented by student Elvis Fefey.
- October, 2022 **Performance Modeling Across Heterogeneous Domains Using Few-Shot Learning**, *New York Scientific Data Summit (NYSDS)*, Virtual, Lightning talk presented by student Arunavo Dey.
- October, 2022 **Analysis and Visualization of Important Performance Counters to Enhance Interpretability of Auto-tuner Output**, *New York Scientific Data Summit (NYSDS)*, Virtual, Lightning talk presented by student Mohammad Zaeed.
- September, 2022 **How to Apply to Graduate School: Things I Wish I Knew**, *Bangladesh University of Engineering & Technology (BUET)*, Virtual.
- May, 2022 **How to be a Great Mentor**, *Exascale Computing Project (ECP)*, Virtual.
- April, 2022 **Scalability Challenges and Opportunities for I/O Bound Applications**, *CHEOPS Workshop at EuroSys*, Virtual.
- March, 2022 **Exascale Computing Project (ECP): Enabling Next-Generation of Hardware-Software Co-design using Data Science**, *Lawrence Livermore National Laboratory (LLNL)*, Virtual.
- December, 2021 **Characterizing Performance of Workflow Applications**, *Brookhaven National Laboratory (BNL)*, Virtual.
- December, 2021 **Center of Excellence: Performance Characterization of Deep Learning Workloads**, *AMD*, Austin, TX.
- September, 2021 **Enabling Next-Generation Software and Co-Design**, *AMD at Texas State University*, San Marcos, TX.
- September, 2021 **HPC Enables Digitalization Panel**, *Digital 360 Summit*, San Marcos, TX.
- November, 2020 **Careers in HPC Panel—Taking the Leap: Changing Careers**, *International Conference for High Performance Computing, Networking, Storage and Analysis (SC)*, Virtual.
- October, 2020 **May the HPC Force be with You!**, *Richard Tapia Celebration of Diversity in Computing*, Virtual, Organized a panel introducing undergraduate students from underserved and underrepresented communities to HPC.
- August, 2020 **Data-Driven Performance Modeling for HPC Co-design**, *AMD*, Virtual, Invited scholar's talk.
- August, 2020 **Dashing: An Interpretable Machine Learning Toolkit for Performance Analysis and Visualization**, *Exascale Computing Project Hackathon at Oak Ridge National Laboratory and Brookhaven National Laboratory*, Virtual.
- May, 2019 **Scalable I/O Performance for Scientific Applications—Challenges and Potentials**, *NIMBioS Workshop on Scientific Collaboration Enabled by High Performance Computing*, Knoxville, TN.
- December, 2018 **Understanding the Performance Portability Challenges and Opportunities using Machine Learning**, *Lawrence Berkeley National Laboratory (LBNL)*, Berkeley, CA.
- November, 2018 **HPC for Undergraduates**, *International Conference for High Performance Computing, Networking, Storage and Analysis (SC)*, Dallas, TX.
- July, 2018 **Machine Learning Framework for Performance Coverage Analysis of Proxy Applications**, *The Platform for Advanced Scientific Computing Conference (PASC)*, *ACM and the Swiss National Supercomputing Centre (CSCS)*, Basel, Switzerland.
- November, 2015 **How HPC Shaped My Career**, *International Conference for High Performance Computing, Networking, Storage and Analysis (SC)*, Austin, TX.
- February, 2015 **Proxy Application Validation using Machine Learning (Veritas)**, *JOWOG-34*, Sandia National Laboratories, Albuquerque, NM.
- January, 2014 **Opportunities after Graduate School**, *Spelman College*, Atlanta, GA.

Teaching Experience

- 2021–2026 HPC@Scale, *Spring*
2021–2025 Linear Data Structures (CS-II), *Fall 2021, 2023, 2025*
2020–2024 Compiler Construction, *Spring 2020; Fall 2020–2022, 2024*
2022 Data Science Pathways (online certification program), *Summer*
2018–2019 Graduate/Undergraduate Parallel Computing (WWU), *Winter, Fall, Spring*
2018–2019 Computer Networks (WWU), *Spring, Winter*
2017–2019 Advanced Data Structures (WWU), *Fall, Spring, Winter*
2018 Scientific Data Visualization (WWU, co-designed and co-taught), *Fall*

Outreach and Mentoring Students

- 2025–Present **Coach**, *FIRST LEGO League, elementary-school team*, Central Texas, Best Innovation Project Award, Central Texas Qualifier (2026); Best Robot Design Award, Texas Regional Championship (2026). Volunteer coach teaching children to design, build, and program robots and to research solutions to real-world problems.
- 2014–Present **Co-founder**, *Bangladeshi Women in Computer Science and Engineering*, Dhaka, Bangladesh, First research, networking, and mentoring platform for female Computer Science and Engineering students of Bangladesh. Mentored 100+ female students through graduate applications, fellowship and scholarship submissions, research and internship preparation, career planning, and leadership development in computing.
- 2017–Present **Research advisor**, *Undergraduate (10), M.S. (5), Ph.D. (8), and Postdoc (2) (TXST); Undergraduate (12) and graduate (7) students (WWU)*, High school (2); K-5 AI (4).
- 2018–Present **Student internships**, *National laboratories: LLNL, LBNL, BNL, ORNL, PNNL*. Nirajan Banjade (Quantum Science Center, ORNL, 2026); Banooqa Banday (BNL, 2023; LBNL, 2026); Aakash Dhakal (LLNL, 2024, 2025); Ashna Nawar Ahmed (ORNL, 2024); Chase Phelps (LLNL, 2021; BNL, 2022, 2023); Jeffrey Boerger (BNL, 2023); Ankur Lahiry (BNL, 2022); Alicia Guite (BNL, 2022); Mohammad Zaeed (LBNL, 2022); Alexis Ayala (LBNL, 2019); Quentin Jensen (LBNL, 2019); Holland Schutte (LLNL, 2019); Nathan Pinnow (LLNL, 2018); Gian-Carlo DeFazio (LLNL, 2018).
- 2019–Present **Mentor**, *REU (3), Graduate students (10)*, SC Student Volunteer Program.
- 2017–2019 **Faculty advisor**, *ACM-W chapter at WWU*, ACM-ICPC Programming Club, Faculty mentor.
- 2014–2016 **Intern supervisor**, *Undergraduate, M.S. and Ph.D. students from University of Hamburg, University of Illinois Urbana-Champaign, Spelman College, University of California San Diego*, NSF REU Advisor, Franklin & Marshall College, University of California Berkeley.
- 2011–2013 **Student mentor**, *Undergraduate, M.S., Ph.D. students*, Purdue University.

Press Articles and Media Recognition

- September, 2026 **ACM SIGHPC Names Tanzima Z. Islam and Olga Pearce 2026 Emerging Woman Leader in Technical Computing**, *ACM SIGHPC*, also covered by the TXST Department of Computer Science ([link](#)).
- September, 2026 **Dr. Islam—Co-General Chair of upcoming IEEE Conference**, *TXST Department of Computer Science News*, IPDPS 2027; also announced by the Sustainable Horizons Institute ([link](#)).
- July, 2026 **TXST faculty lead Department of Energy Genesis Mission projects to advance efficient, reliable science**, *Texas State University Newsroom*, also listed by Brookhaven National Laboratory ([link](#)) and in the DOE Genesis Mission awards list ([link](#)).
- April, 2025 **TXST's Tanzima Islam awarded NSF CAREER grant to improve supercomputer efficiency**, *Texas State University Newsroom*, also covered by the TXST Department of Computer Science ([link](#)).
- February, 2025 **Embracing artificial intelligence could elevate TXST**, *The University Star, Texas State University*, quoted on TXST's position in AI.
- February, 2024 **SRP at the DOE Advanced Scientific Computing Research Computer Science Principal Investigator (ASCR CS PI) Meeting**, *Sustainable Horizons Institute*.
- November, 2023 **CSI in the Sky: Brookhaven Lab Heads to SC23 in Denver**, *Brookhaven National Laboratory Newsroom*, features SC23 poster by students Chase Phelps and Ankur Lahiry.
- November, 2023 **Presidential Award Winners 2023**, *Texas State University, Office of the President*.

- May, 2023 [Berkeley Lab Computing Sciences Takes Novel Approach to Science Workforce Development and Diversity](#), *Berkeley Lab Computing Sciences News*.
- November, 2022 [Brookhaven Scientists Ready to Show how 'HPC Accelerates' at SC22](#), *Brookhaven National Laboratory Newsroom*, features SC22 poster by student Alicia Guite.
- July, 2022 [SRP Paves the Way for DOE Early Career Award](#), *Berkeley Lab Computing Sciences News*, features WWU students Quentin Jensen and Alexis Ayala.
- June, 2022 [Tanzima Islam's innovative computing work garners Early Career Research Program award](#), *Texas State University Newsroom*, also listed in the DOE Office of Science 2022 awards list ([link](#)).
- May, 2022 [Translational Health at Texas State University](#), *AMD (video)*, AMD HPC Fund feature on deep vision for breast cancer classification and segmentation.
- November, 2019 [LLNL researchers win four R&D 100 awards](#), *Lawrence Livermore National Laboratory News*, SCR 2.0 team.
- October, 2010 [Graduate student Tanzima Zerine Islam wins award in ACM Student Research Competition](#), *Purdue University, School of Electrical and Computer Engineering News*.